

EXERCISE 5.2

Equations Reducible to Quadratic Form

(Short Answer Questions 1–5 / Long Answer Questions 6–18)

Short Answer Questions

1. (i)

$$\frac{x-2}{x+2} + \frac{x+2}{x-2} = 4$$

(ii)

$$\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}$$

2.

$$\frac{1}{x+1} - \frac{2}{x+2} = \frac{3}{x+3} - \frac{4}{x+4}$$

3. (i)

$$3^x + 3^{-x} = 2$$

(ii)

$$2^{x+1} + 4^x = 8$$

4.

$$5^{x+1} + 5^{2-x} = 5^3 + 1$$

5.

$$x^{2/3} - x^{1/3} - 2 = 0$$

Long Answer Questions

6. (i)

$$(x^2 + 3x + 2)^2 - 8(x^2 + 3x) - 4 = 0$$

(ii)

$$(x^2 - 5x + 7)^2 - (x-2)(x-3) = 1$$

7.

$$x(x+2)(x+3)(x+5) = 72$$

8.

$$x(x+2)(x^2-1) = -1$$

9.

$$(x+1)(2x+3)(2x+5)(x+3) = 945$$

10.

$$(x^2 + x - 6)(x^2 - 3x - 4) = 24$$

11. (i)

$$\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$$

(ii)

$$\sqrt{\frac{4x-1}{4x+1}} - \sqrt{\frac{4x+1}{4x-1}} = \frac{8}{3}, \quad x \neq \pm \frac{1}{4}$$

12. (i)

$$\sqrt{3x+4} = x$$

(ii)

$$\sqrt{3x+1} - \sqrt{x-1} = 2$$

13. (i)

$$\sqrt{x+1} + \sqrt{x-1} = 1$$

(ii)

$$\sqrt{x+3} + \sqrt{x-3} = 6$$

14. (i)

$$\sqrt{3x^2 - 4x + 34} - \sqrt{3x^2 - 4x - 11} = 5$$

(ii)

$$\sqrt{3x^2 - 7x - 30} - \sqrt{2x^2 - 7x - 5} = x - 5$$

15.

$$12 + 9\sqrt{(x-1)(3x+2)} = 3x^2 - x$$

16.

$$\frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{\sqrt{x^2+1} + \sqrt{x^2-1}} = \frac{1}{3}$$

17. (i)

$$\left(x + \frac{1}{x}\right)^2 - 2\left(x - \frac{1}{x} + 4\right) = 11$$

(ii)

$$12x^4 - 56x^3 + 89x^2 - 56x + 12 = 0$$

18.

$$(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$$

ANSWERS

1. (i) $2\sqrt{3}, -2\sqrt{3}$ (ii) $5, \frac{5}{2}$

2. $0, -\frac{5}{2}$

3. (i) 0 (ii) 1

4. $-1, 2$

5. $-1, 8$

6. (i) $-4, -3, 0, 1$ (ii) $2, 3, \frac{5 \pm i\sqrt{3}}{2}$

7. $1, -6, \frac{-5 \pm i\sqrt{23}}{2}$

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